|QW⟩ group

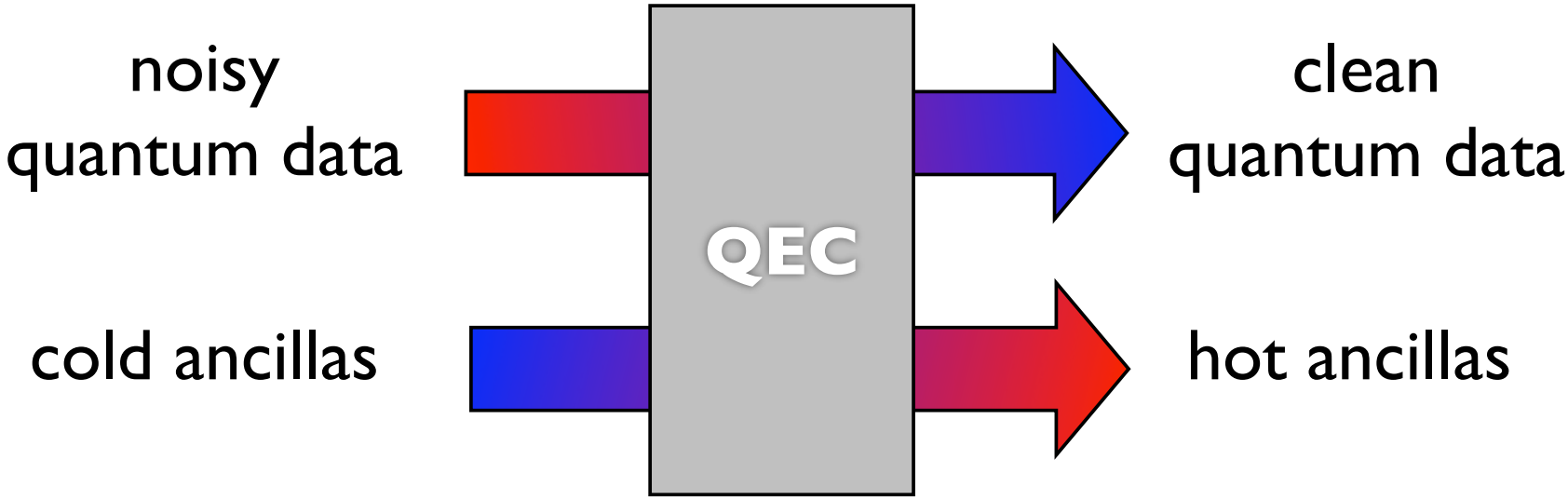


Robust Quantum Gadgets

Principle Investigator: Dave Bacon
Graduate Students: Gregory Crosswhite (Physics)
Alice Neels (CS)
Postdoc: TBD

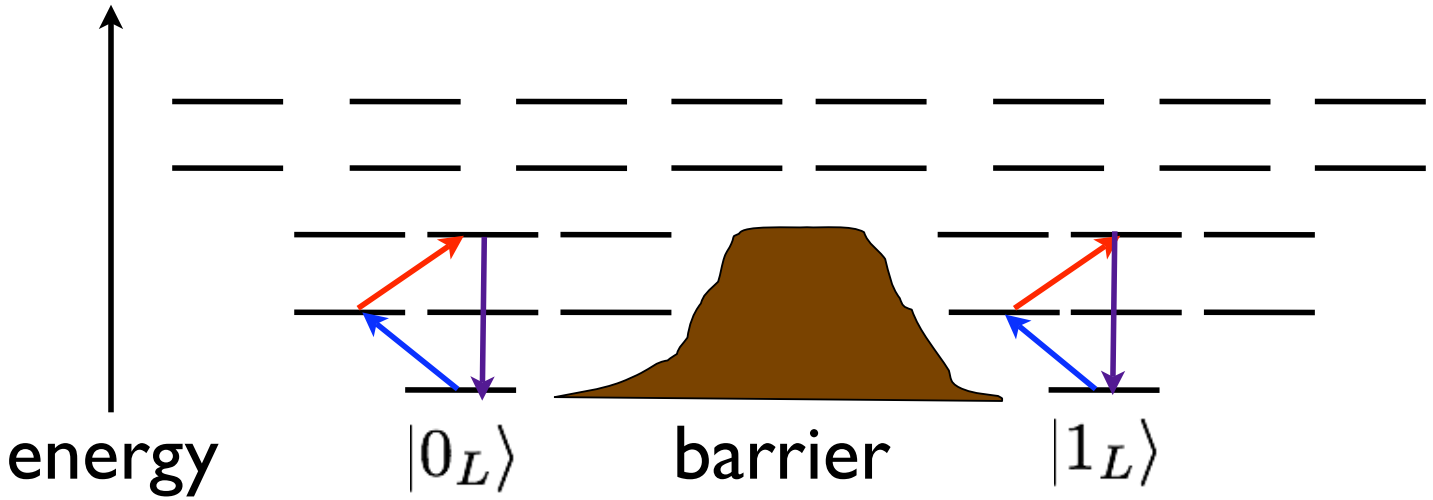
University of Washington
Department of Computer Science & Engineering
Department of Physics

The Physics Approach to Quantum Error Correction



(Quantum) error correction is nothing more than a sophisticated refrigeration process...can we build physical systems where quantum error correction is achieved by simply keeping the system cold?

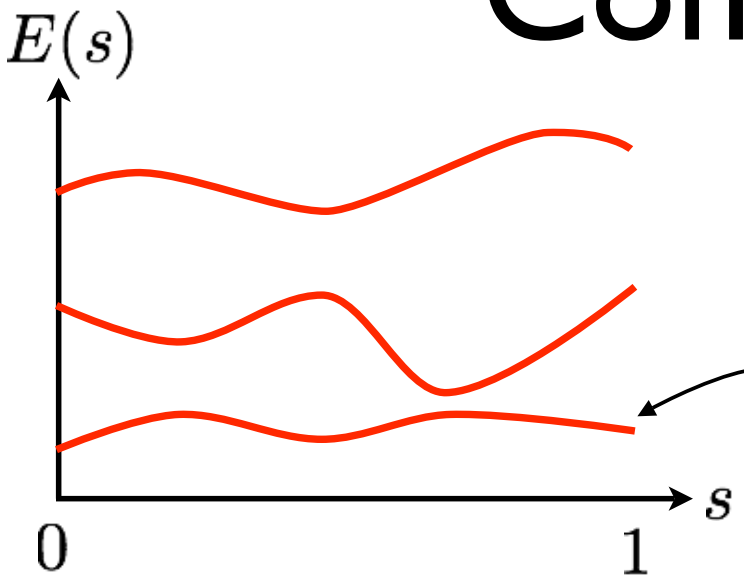
Self-Correcting Quantum Computers



Energy landscapes with quantum error correcting code states can create systems which "self-correct"

Problem: known examples are not physically realistic

Universal Adiabatic Quantum Computers



adiabatic Hamiltonian:

$$H(s) = (1-s)H_0 + sH_f$$

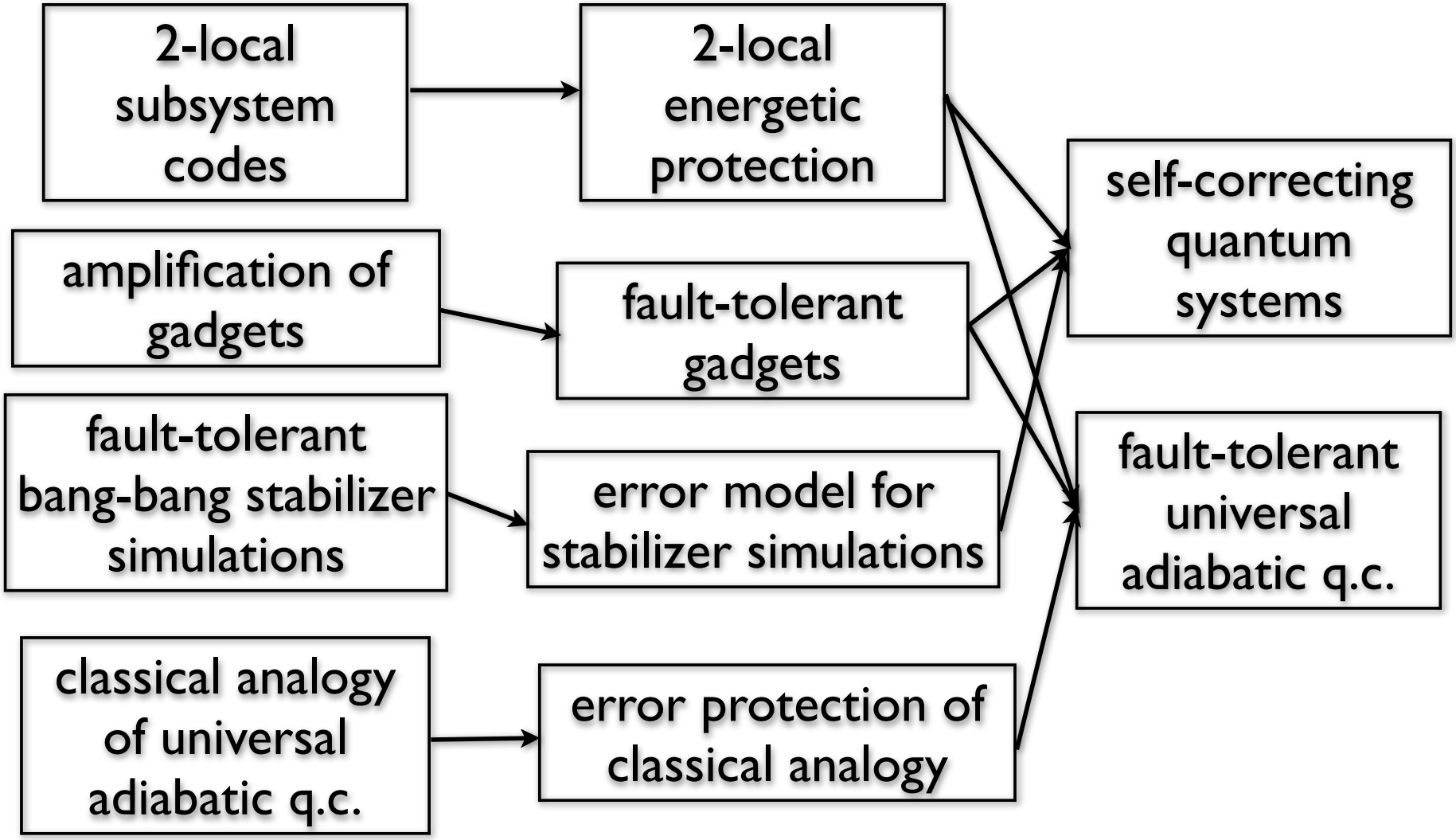
$$\frac{1}{\sqrt{T}} \sum_{t=1}^{T} |t\rangle \otimes U(t)|0\rangle$$

superposition over history of quantum circuit

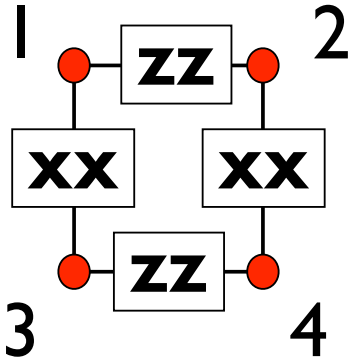
adiabatic quantum algorithms are equivalent to the quantum circuit model

Problem: not known to be fault-tolerant

QuEST Project



Energetic Protection I



$$H = -\Delta(X_1X_3 + X_2X_4 + Z_1Z_2 + Z_3Z_4)$$

[4,2,2] QECC

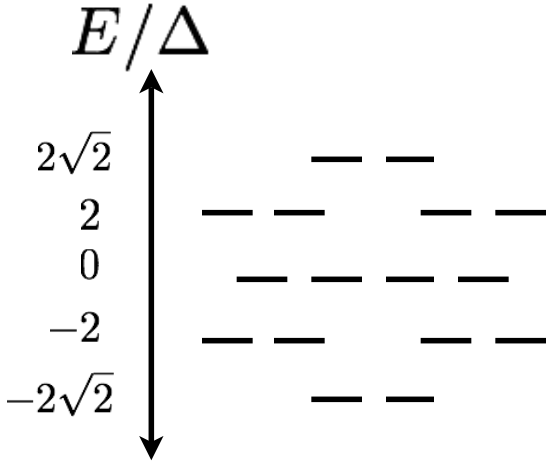
stabilizer

$$S_1 = X_1X_2X_3X_4 \qquad S_2 = Z_1Z_2Z_3Z_4$$

logical qubits

$$\bar{X}_1 = X_1X_3 \quad \bar{Z}_1 = Z_1Z_2$$

$$\bar{X}_2 = X_1X_2 \quad \bar{Z}_2 = Z_1Z_3$$

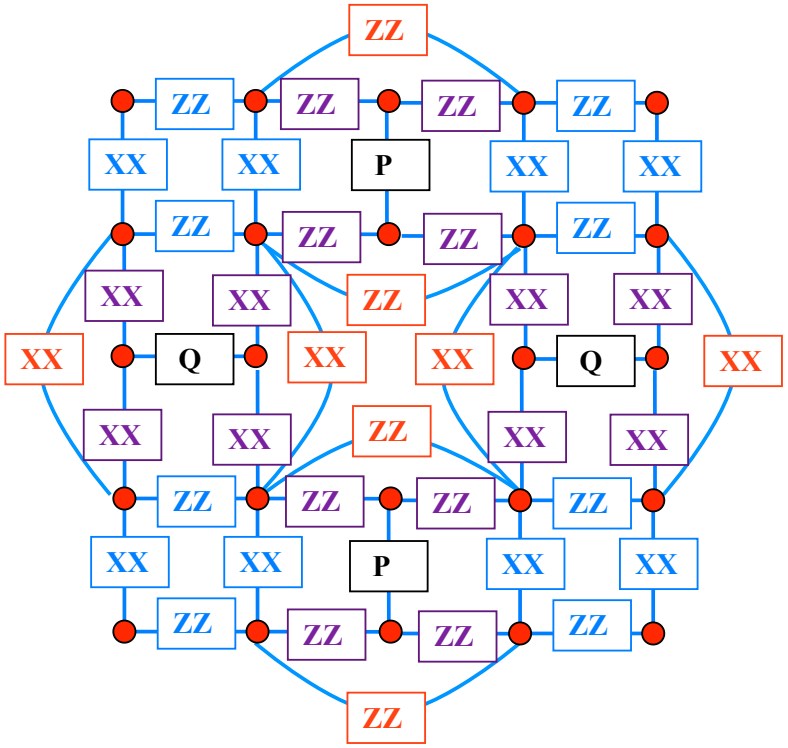


ground states are error detecting code

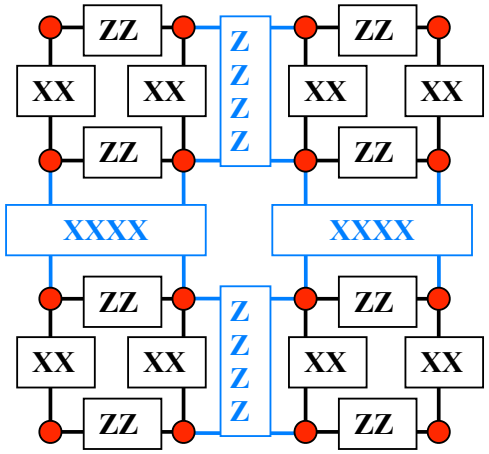
$$H = -\Delta(\bar{X}_1(I + S_1) + \bar{Z}_1(I + S_2))$$

S_1	S_2	H
$+1$	$+1$	$-2\Delta(\bar{X}_1 + \bar{Z}_1)$
-1	$+1$	$-2\Delta\bar{Z}_1$
$+1$	-1	$-2\Delta\bar{X}_1$
-1	-1	0

Energetic Protection II



≅



concatenated [[16,1,4]] code

Concatenated Hamiltonians can provide self-correction, but scale poorly

Subsystem Codes

Most general manner of encoding quantum information is into a subsystem:

$$\mathcal{H} = \bigoplus_s \mathcal{H}_s \otimes \mathcal{J}_s$$

Two known subsystem codes known to be useful in qc:
stabilizer group and symmetric group codes

Subsystems codes are especially important for energetic protection because they allow Hamiltonian with terms which act on dual subsystem...but not many subsystem codes are known

2-Local Subsystem Code Survey

Problem: Classification of subsystem codes arising from 2-local interactions

Example: 2-local Ising type interactions (see other poster)



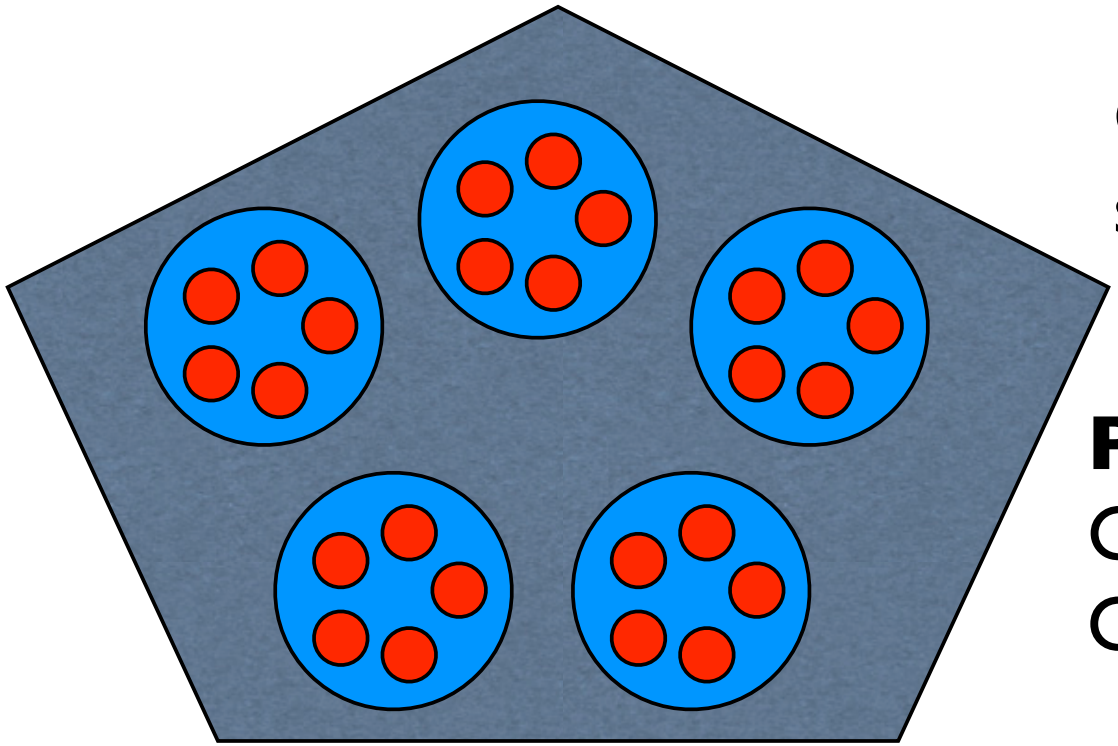
Goal: small scale examples of supercoherent and self-correcting quantum systems with realistic couplings

Stabilizer Hamiltonians

Hamiltonian with sum of stabilizer generators:

$$H = -\Delta(XZZXI + IXZZX + XIXZZ + ZXIXZ)$$

Ground state is stabilizer code.

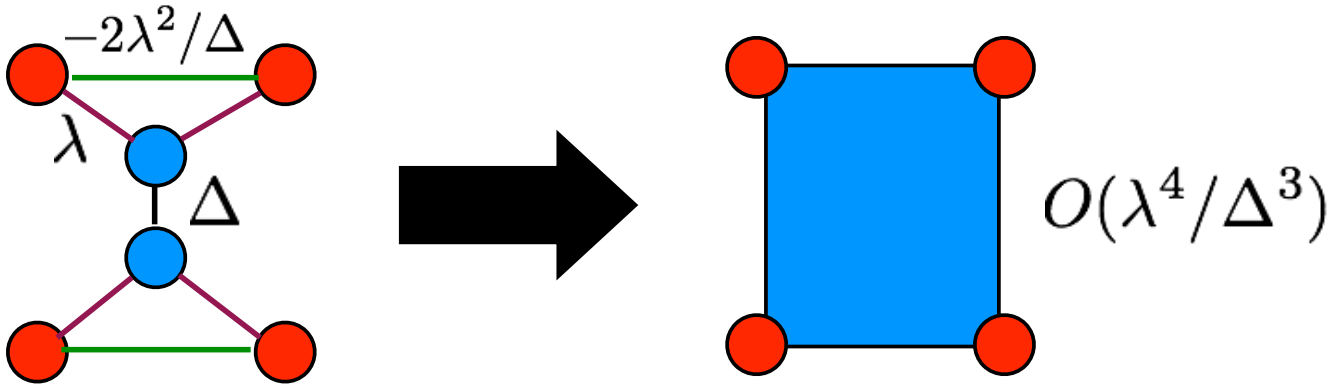


Concatenate to get self-correction

Problem:
Concatenation leads to O(n)-local interactions

Perturbation Gadgets

Create O(k)-local interactions using perturbative gadgets

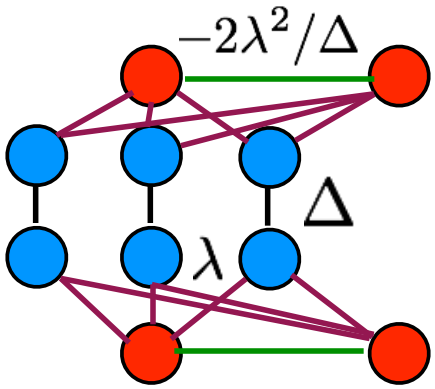


2-local interactions

4-local interaction

Problem: strength of interactions scales exponentially in the size of the desired interaction

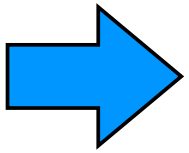
Amplifying Gadgets



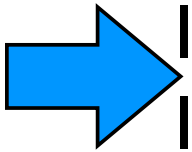
Use multiple ancillas to amplify strength....but this just replaces problem with # ancillas growing exponentially

Overcome this by "concatenating" gadgets.

100 2-local
1 3-local



100 3-local
1 6-local

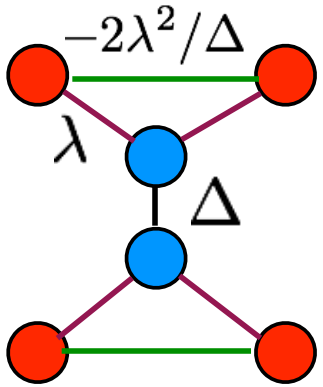


100 6-local
1 15-local

10^k 2-local
1 $(3^{k/2+1}-3)/2$-local

Fault-Tolerant Gadgets

Problem: Gadgets themselves are not robust to noise on the system that is creating them



must analyze errors on ancillas

Alternatively: ancillas which are themselves energetically protected?

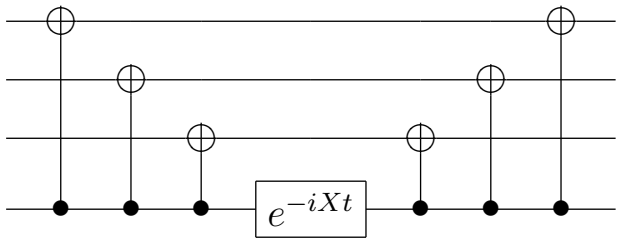
Bang-Bang Simulation

Do we retain robustness of self-correction if we simulate it in the circuit model?

Trotterization can be used to break down terms,

$$\exp(i(H_1 + H_2)t) = \lim_{N\to\infty} \prod_{i=1}^{N} \exp(iH_1 t/N) \exp(iH_2 t/N)$$

and standard conjugation tricks used to create O(n)-local interactions

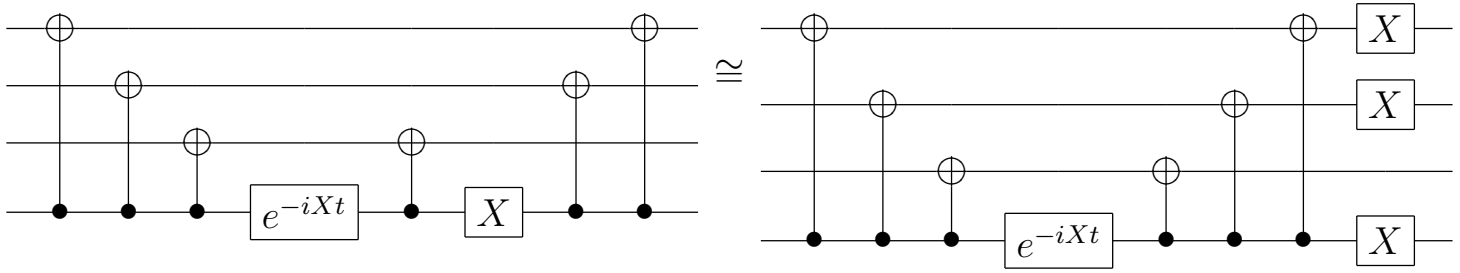


$$\cong \quad \exp(-iXXXXt)$$

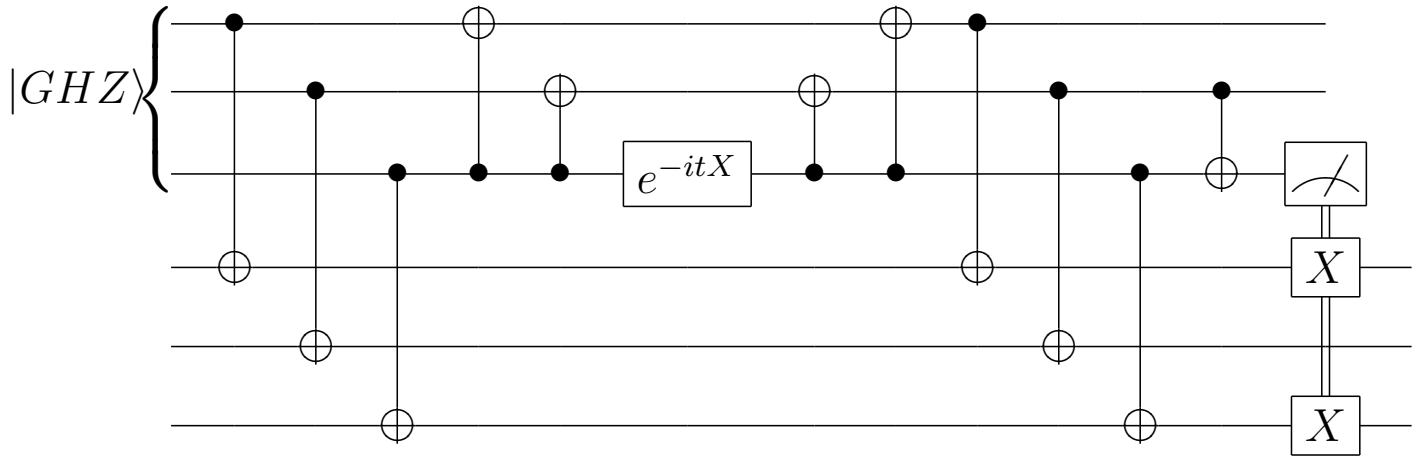
but this is not fault-tolerant!

Fault-Tolerant Bang-Bang

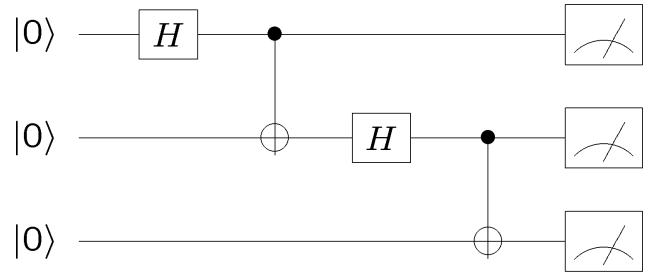
The problem:



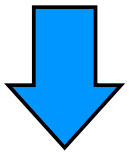
Fault-tolerant stabilizer simulation (don't care about stabilizer errors):



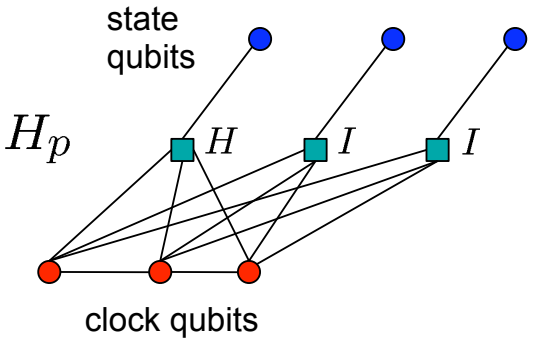
Universal Adiabatic QC



Quantum Circuit



Hamiltonian with history of circuit as ground state



$$|\psi_0\rangle = \frac{1}{\sqrt{s+1}} \sum_{i=0}^{s} |\psi(i)\rangle \otimes |i\rangle_{clock}$$

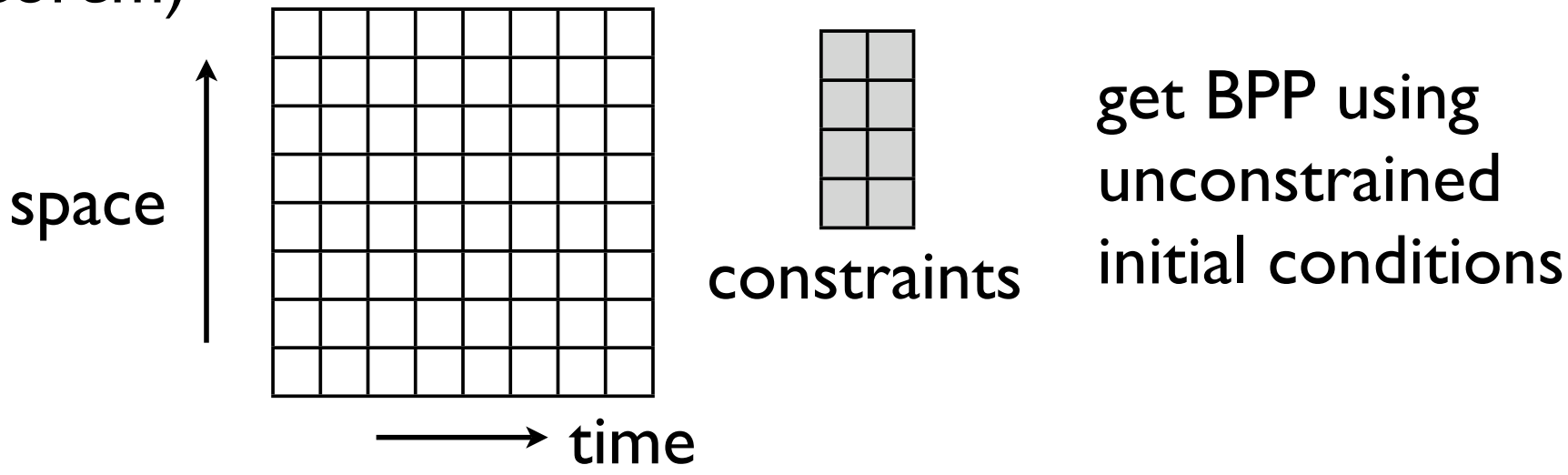
$$|\psi(i)\rangle = U_i U_{i-1} \cdots U_1 |\psi(0)\rangle$$

Possible to adiabatically drag oneself to this ground state in polynomial time

Classical Analogy of Universal Adiabatic QC

Problem: Fault-tolerance of universal AQC not understood

Construct classical analogy and study it's robustness (intermediary between universal AQC and Cook-Levin theorem)



References



D. Bacon, K.R. Brown, and K.B. Whaley, "Coherence-Preserving Quantum Bits," Physical Review Letters, 87, 247902-1-247902-4 (2001)

D. Bacon, "Operator Quantum Error-correcting Subsystems for Self-correcting Quantum Memories," Physical Review A, 73, 012340 (2006)

D. Bacon, "Stability of Quantum Concatenated Code Hamiltonians," Phys. Rev. A, 78, 042324 (2008)